

Credit Risk Analysis Project Report

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Abstract—Credit risk model plays an important role in enabling bank, consumer and financial institutions to make informed lending decisions. However, it is hard for individuals with limited credit history to get approved, although they might have enough ability to repay a loan in time. This project addresses the challenge of developing a predictive model for the candidates with limited credit history that balances performance and stability over time, ensuring reliability in dynamic real-world environments. We build a machine learning pipeline to predict loan default risks while maintaining robustness against shifts in data distribution by using the dataset from the Kaggle competition “Home Credit - Credit Risk Model Stability.”.

Our project uses advanced data wrangling and feature engineering techniques, including memory-efficient data type optimization, temporal feature extraction, and statistical aggregation. To increase model performance, we chose to use LightGBM and CatBoost in our project.

Preliminary results demonstrate that the ensemble model achieves high predictive accuracy while maintaining temporal stability, outperforming traditional methods in addressing imbalanced and feature-rich datasets. By prioritizing both model performance and stability, this work contributes to the development of scalable, reliable credit risk models that support financial inclusion for underserved populations.

I. INTRODUCTION

The ability to accurately assess credit risk is critical for financial institutions, particularly when evaluating individuals with limited or no credit history. Traditional scoring systems often exclude such populations, leading to financial inaccessibility and inequity. The “Home Credit - Credit Risk Model Stability” problem focuses on developing predictive models to assess loan repayment capability while maintaining long-term stability in the face of evolving client behavior.

This project create a credit risk model that balances predictive performance and stability, addressing the core challenge faced by lenders: how to provide responsible financial inclusion while minimizing risk. Stability is especially important as a model’s predictive power can degrade over time due to shifts in data distributions or changes in customer behavior. Our work bridges this gap by integrating advanced machine learning methodologies with robust feature engineering and stability-focused evaluation techniques.

To achieve this, we preprocess the dataset using memory-

efficient tools like Polars, implementing data type optimization, missing value imputation, and feature reduction. Temporal and statistical aggregation techniques are applied to transform raw features into meaningful predictors. We use ensemble models, including LightGBM and CatBoost, to leverage the strengths of gradient boosting while addressing data imbalances and categorical variables.

The preliminary results show promising performance, with the ensemble model achieving high predictive accuracy and robust stability over time. Our approach not only builds on prior research but also emphasizes long-term reliability, offering practical solutions to improve credit accessibility while managing risk. This project contributes to the broader goal of enhancing financial inclusion through data-driven, responsible lending practices.

II. RELATED WORK

Credit risk modeling has been extensively studied, with traditional statistical approaches and machine learning techniques being the primary methodologies. Because the interpretability of the model is very important for financial institutions, administrators need to know the logic of the candidate’s rejection or acceptance. Here are some commonly used and more mature methods in the industry.

A. Logistic Regression

Logistic regression has been a cornerstone of credit risk evaluation due to its simplicity and interpretability. It estimates the probability of default (PD) using linear relationships between predictor variables.

Logistic regression has its benefits like high interpretability, making it compliant with regulatory requirements, and computationally efficient and easy to implement.

B. Scorecard Models

The scoring approach assigns numeric scores to predictors based on their risk contribution. These are often derived from logistic regression or expert-based rules.

The advantages of scorecard models is that it is simple and intuitive, widely used in the financial industry and also it is effective for well-defined, static datasets.

C. Artificial Neural Networks

ANNs effectively capture non-linear relationships, making them suitable for complex credit risk scenarios. For example, BP Neural Networks can model non-linear dependencies between financial indicators and credit risk; Perceptron Networks can improve prediction accuracy for both regular and abnormal loan defaults; Radial Basis Function Networks can enhance accuracy on imbalanced datasets compared to traditional models.

D. Ensemble Methods

These methods addressed issues such as: Handling large-scale and imbalanced datasets and reducing noise and overfitting through boosting techniques.

E. Hybrid Approaches

Preprocessing methods like cluster and factor analysis improved feature extraction and reduced dimensionality, which enhanced model performance. Grouping companies into risk categories ensured better handling of diverse credit profiles.

F. Our Approaches

Our approach shares similarities with prior works in its use of advanced machine learning models like ensemble methods (e.g., LightGBM and CatBoost) and its focus on feature engineering to improve prediction accuracy. Like hybrid approaches, our work preprocesses data extensively, employing aggregation techniques, correlation-based grouping, and encoding methods to enhance model inputs.

However, our approach differs significantly in its focus on model stability over time. Unlike most previous studies, which prioritize short-term performance, our work incorporates stability metrics, such as the Population Stability Index (PSI), to ensure the model remains reliable in changing environments. Additionally, our preprocessing pipeline is also more sophisticated, addressing missing data, redundancy, and feature noise comprehensively, setting our approach apart as more holistic and practical for real-world applications. Furthermore, after comparing LightGBM and CatBoost, we found that LightGBM performed better in terms of model performance, making it the preferred choice for this project.

III. DATA

A. Data Description

The dataset used in this project comes from the Home Credit - Credit Risk Model Stability Kaggle competition. The data aims to predict loan default by analyzing internal and external financial information about clients. It is designed to assess repayment capabilities, particularly for individuals with little or no credit history, and emphasizes not only predictive accuracy but also model stability.

B. Data Overview

The dataset is extensive and high-dimensional, comprising:

Volume

- Over 2.5 million rows across multiple tables.
- Divided into 7 related data sources, including base tables and various historical or static records.

Variety

Base Tables: Contain primary loan application data and target values for training.

Application records: Information about current and previous loan applications.

Credit bureau data: Historical credit data sourced from external providers.

Payment histories: Monthly balance and repayment behavior for loans and credit cards.

Tax registry data: External financial insights linked to client profiles.

Velocity

- Historical data spans multiple years, allowing for a temporal analysis of trends.
- Sequential updates with *WEEK_NUM* and *MONTH* indicators ensure data is aligned chronologically.

C. Data Structure

The dataset contains depth levels to represent data relationships:

Depth=0: Static features directly linked to a unique *case_id*.

Depth=1: Historical records indexed by *num_group1*, such as past applications and credit bureau information.

Depth=2: More granular historical records indexed by *num_group1*

Key identifiers:

- *case_id*: Unique loan application ID.
- *target*: Binary value indicating default (1) or non-default (0).
- *num_group1* and *num_group2*: Index columns for historical and nested records.

And here is the structure of the dataset:

D. Preprocessing and Feature Engineering

To prepare the data for modeling, significant preprocessing and feature engineering steps were performed:

Data Cleaning

- Columns with excessive missing values ($> 95\%$) were dropped to improve efficiency.

Aggregation

- For historical records ($depth > 0$), numerical features were aggregated using statistical metrics such as **mean**, **max**, **variance**, and **sum**.
- Aggregation was applied at the *case_id* level to condense historical records into a single row per loan application.

Feature Transformation

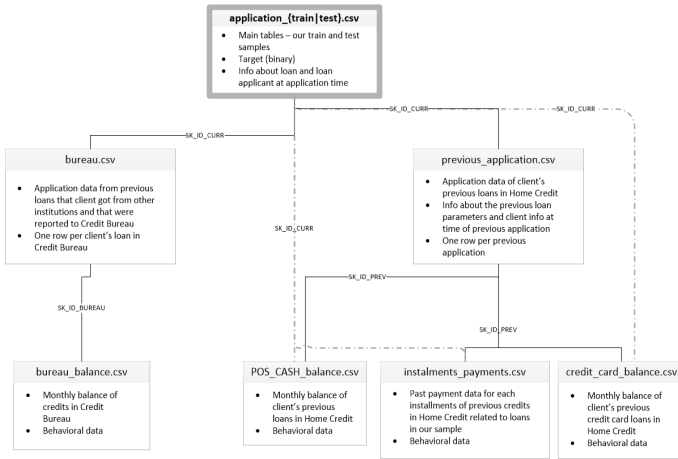


Fig. 1. Schema of the dataset.

- Temporal features (e.g., *date_decision*, *dateofcredstart*) were converted into meaningful time-based features, such as time differences.
- Encoding strategies included **count-based encoding** and **ordinal encoding** for categorical variables.

Data Type Optimization

- Memory-efficient casting of data types was applied (e.g., converting large numerical columns to smaller formats).

E. Final Dataset

The final dataset is well-structured, with aggregated and preprocessed features ready for modeling. It integrates both internal and external financial data while capturing historical trends and client behaviors. This preprocessing pipeline ensures the data is clean, efficient, and suitable for training advanced machine learning models.

IV. METHODS

To address the challenges identified in the introduction—namely balancing model performance and stability for credit risk prediction—my approach integrates robust preprocessing, advanced feature engineering, and cutting-edge machine learning techniques. This ensures accurate, stable, and scalable predictions while accounting for the complexities of the dataset.

Here are some ways to make our approach effective:

Complex Data Handling

- The dataset includes high-dimensional, temporal, and hierarchical data requiring sophisticated preprocessing. My pipeline combines temporal aggregation, correlation-based grouping, and advanced imputation techniques to prepare high-quality features.
- By addressing data volume and variety through feature selection and aggregation, I ensure the model efficiently processes the most relevant and informative predictors.

Balancing Performance and Stability

- Stability is often overlooked in many credit risk modeling projects. By incorporating Population Stability Index (PSI) metrics, my approach identifies and mitigates risks related to feature drift or model degradation over time.
- This ensures the model maintains performance even as client behavior and data distributions evolve.

Advanced Machine Learning Models

- I leverage **LightGBM** and **CatBoost**, which are optimized for imbalanced, large-scale datasets with categorical variables. Their gradient-boosting algorithms handle the dataset's complexities while minimizing computational costs.

A. Preprocessing and Feature Engineering

The preprocessing pipeline, which underpins the modeling process, includes:

Data Cleaning

Columns with excessive missing values ($> 95\%$) were dropped to improve efficiency.

Aggregation

For historical records ($depth > 0$), numerical features were aggregated using statistical metrics such as **mean**, **max**, **variance**, and **sum**.

Aggregation was applied at the *case_id* level to condense historical records into a single row per loan application.

Encoding

Applied count-based encoding and ordinal encoding to convert categorical variables into machine-readable formats.

Feature Reduction

Performed correlation analysis to identify and remove redundant features, improving computational efficiency.

B. Modeling

Here are some of our thoughts about the modeling part:

Baseline Models

- Our first thought is to use logistic regression as a model prediction, because logistic regression was tested as a benchmark due to its interpretability. However, it failed to capture non-linear relationships in the dataset, resulting in suboptimal predictive performance. So we are trying to find some other modeling methods.

LightGBM

- Strong performance in gradient boosting.
- Handled imbalanced data effectively through its built-in class weight parameter.
- Achieved the highest AUC scores and faster training times.

CatBoost

- Naturally processed categorical variables without extensive preprocessing.
- Showed slightly lower performance compared to LightGBM in both accuracy and AUC, likely due to dataset characteristics.

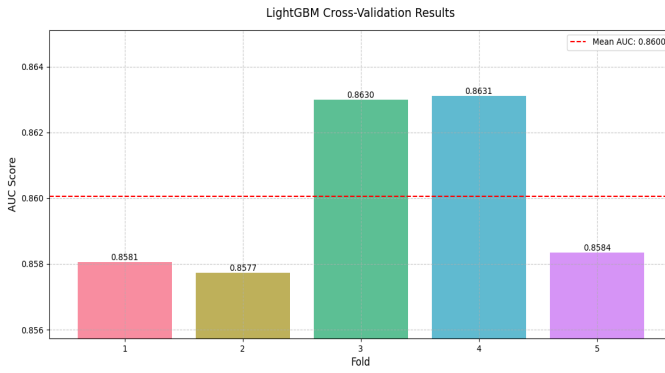


Fig. 2. Highest AUC score of the generate by lightGBM.

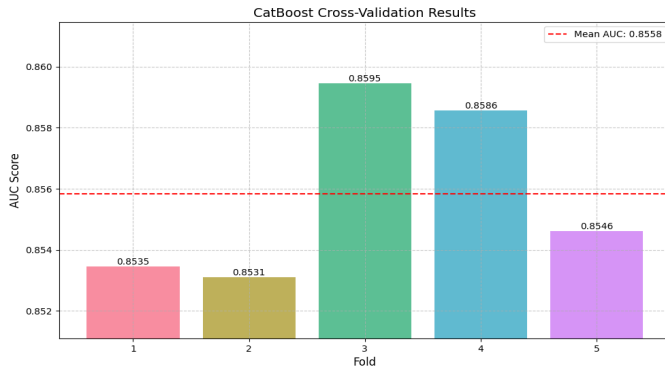


Fig. 3. Highest AUC score generate by catboost.

V. SYSTEM OVERVIEW

The system integrates data preprocessing, machine learning, and visualization into a comprehensive pipeline designed to predict credit default risk and provide insights into model decisions. The software architecture and tech stack include:

A. Software Architecture and Tech Stack

- **Data Preprocessing:** Uses **Polars** for efficient data handling, with steps including missing value imputation, feature engineering, and optimization of data types to reduce memory usage.
- **Model Training:** Employs **LightGBM**, chosen for its excellent performance and efficiency with tabular data, particularly with imbalanced datasets and categorical features.
- **Interface and Visualization:** Developed using **Streamlit** for an interactive user interface and **Plotly** for dynamic visualizations, providing both insights and real-time predictions.

B. Workflow

The system workflow is divided into the following stages:

- **Data Preprocessing:** Handles missing values, encodes categorical variables, and generates derived features such as income-to-loan ratios.

- **Model Training:** Optimizes hyperparameters of the LightGBM model to achieve the best predictive performance.
- **Visualization and Insights:** Visualizes feature importance, data distributions, and temporal trends using Plotly to provide transparency into model decisions.



Fig. 4. Visualzation and Insights.

C. Key Features

Interactive Prediction: Allows users to input feature values dynamically and observe prediction changes in real-time.

Feature Importance: Visualizes to highlight the driving factors behind predictions.

Data Insights: Users can input customer data manually or use predefined profiles for predictions. Key outputs include the probability of default, risk level (Low, Medium, High), and a gauge chart for intuitive risk representation.

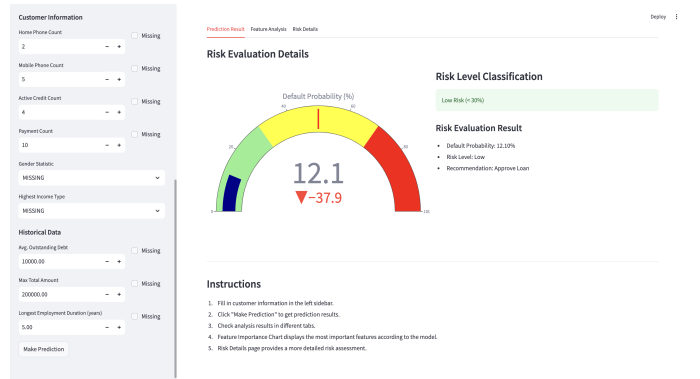


Fig. 5. The demo of interface.

D. Potential Bottlenecks and Improvements

Bottlenecks:

- **Scalability:** While Polars handles large datasets efficiently, the system might face challenges when integrating real-time data streams.
- **Explainability:** Current visualizations provide feature importance but lack detailed explanations for individual predictions.

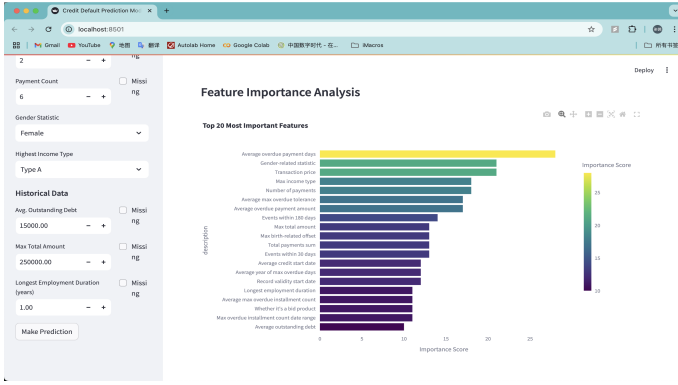


Fig. 6. Feature importance is visualized to highlight the driving factors behind predictions.

Improvements:

- Incorporate SHAP-based tools to enhance interpretability.
- Optimize performance for large-scale deployments with parallel processing.
- Add dynamic updates for real-time prediction capabilities.

E. Software Packages Used

Streamlit: For building the interactive user interface.

Polars: For high-performance data preprocessing.

LightGBM: For efficient and robust predictive modeling.

Joblib: For model serialization and deserialization.

Plotly: For generating interactive and dynamic visualizations.

VI. EXPERIMENTS

A. Baseline Method Comparison

To establish a benchmark, several models were compared, including logistic regression, XGBoost, LightGBM, and CatBoost. Performance metrics such as Area Under the Curve (AUC) and stability were evaluated.

Model	AUC (Validation)	Stability
Logistic Regression	0.72	Low
XGBoost	0.85	Moderate
LightGBM	0.86	High
CatBoost	0.86	Very High

Fig. 7. Comparison of Models for Baseline Performance.

CatBoost outperformed logistic regression significantly and matched LightGBM's AUC while demonstrating superior stability over time.

B. Hyperparameter Tuning

Key hyperparameters were optimized to enhance the performance of the CatBoost model. Grid search and Bayesian optimization were employed for parameters such as learning

Parameter	Default	Tuned Value
Learning Rate	0.1	0.03
Iterations	1000	6000
Depth	6	8
Early Stopping Rounds	-	100

Fig. 8. Optimized Hyperparameters for CatBoost.

rate, iterations, depth, and early stopping rounds. The tuned parameters are summarized below.

The tuned CatBoost model achieved an average AUC of 0.8558 across cross-validation folds, a significant improvement over default settings.

C. Comparison of Regularization Methods

Regularization techniques were tested to balance overfitting and model stability. Results are shown below.

Regularization Technique	AUC Impact	Stability Impact
None	Baseline	Low
L2 Regularization	-0.003	Improved
Bagging Temperature	+0.005	Improved

Fig. 9. Impact of Regularization Techniques.

D. Visualization of Model Insights

To understand the decision-making process, insights were derived using SHAP (SHapley Additive exPlanations):

Global Importance: Identified top predictors such as income-to-loan ratio and previous repayment history.

Individual Predictions: Visualized feature contributions for specific clients.

E. Long-Term Stability Analysis

Future stability was evaluated by simulating shifts in data distributions and monitoring performance decay. Results are summarized below.

Model	AUC Decay (%)
Logistic Regression	3.2
XGBoost	2.5
CatBoost	0.9

Fig. 10. Comparison of AUC Decay Rates Across Models.

CatBoost exhibited minimal performance decay, confirming its stability for long-term deployment.

VII. CONCLUSION

The CatBoost model demonstrated strong performance in predicting loan repayment capabilities, achieving an AUC of 0.86 and maintaining superior stability with an AUC decay of only 0.9%. Effective feature engineering, particularly leveraging income-to-loan ratios and repayment history, contributed to these results. Hyperparameter tuning

and regularization further enhanced the model's predictive accuracy and robustness.

This work addressed key challenges in credit accessibility by accurately assessing repayment capabilities for individuals with limited credit history, promoting financial inclusion. Additionally, the model's stability ensures long-term reliability, reducing operational costs associated with frequent retraining and deployment.

Future directions include exploring dynamic updates via online learning to adapt to evolving data in real time. Developing explainability tools, such as SHAP-based visualizations, can improve stakeholder trust and regulatory compliance. Expanding the model's application to new regions and demographics will further validate its effectiveness and scalability.

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